Business Plan

FreshCart Plus Subscription Clustering Assignment
Business Name: FreshCart Plus

Purpose: Use this business plan with the FreshCart Plus subscription dataset to apply, evaluate, and interpret a K-Means Clustering unsupervised learning model.

1. Business Overview

Item	Description
Business Type	Subscription-based online grocery delivery service
Main Service	Delivering groceries, household essentials, prepared meals, and household products to urban, suburban, and small-town customers through a mobile app and website
Main Customers	Urban households, suburban families, small-town residents, and individual consumers who value convenience in grocery shopping
Main Business Goal	Reduce customer churn by identifying distinct subscriber segments and delivering targeted retention strategies before renewal dates
Machine Learning Goal	Discover natural customer groupings using K-Means Clustering based on subscription behavior, app engagement, delivery experience, and service history

FreshCart Plus is a subscription-based online grocery delivery service that depends on recurring membership revenue. The company currently sends generic renewal reminders and discount offers to all customers. This approach is inefficient because different customers churn for different reasons. The business wants to use unsupervised machine learning to discover which customer groups exist within its subscriber base so that retention strategies can be tailored to each group.

2. Business Problem

The key business problem is undifferentiated customer retention. FreshCart Plus treats most subscribers similarly, regardless of their engagement level, service experience, or spending behavior. Customers leave for different reasons, but the business cannot address those reasons without first identifying distinct customer groups. A single retention strategy wastes budget and misses the customers who need specific attention.

- Some customers churn because of low app engagement and poor perceived value, not service failures.
- Some highly engaged customers are being lost due to poor delivery experiences that could be corrected.
- Some customers face payment barriers that prevent renewal even though they want to stay.
- The marketing team needs a data-driven segmentation model that explains which customers belong to each group and why.
- Managers need actionable cluster profiles that connect directly to targeted retention campaigns.

3. Machine Learning Decision

Machine learning can help the business discover hidden customer groups without requiring predefined labels. K-Means Clustering can segment customers based on their behavioral and service data. Managers can then use these segments to design differentiated retention strategies for each group.

Recommended use of the model

- Identify distinct customer segments based on subscription behavior, engagement, and service experience.
- Support targeted marketing by matching the right retention message to each customer group.
- Understand which behavioral patterns are associated with high renewal rates versus high churn risk.
- Give managers an interpretable segmentation tool that connects data patterns to business actions.

Important note: The model should support human decision-making. It should not automatically replace marketing judgment, customer service judgment, privacy review, or ethical review.

4. Dataset Description

The dataset contains customer-level subscription and behavioral data. Each row represents one FreshCart Plus subscriber. The dataset is used for unsupervised clustering — there is no target variable used during model training, but Renewed_Membership can be used after clustering to evaluate how well the segments correspond to renewal outcomes.

Dataset Element	Explanation
Observation Unit	One FreshCart Plus subscriber record
Target Variable	Renewed_Membership (used for post-hoc evaluation only, not as a clustering input)
Target Meaning	Yes = subscriber renewed the membership; No = subscriber did not renew
Main Feature Types	Customer demographics, membership type, ordering behavior, app engagement, delivery quality, service complaints, competitor exposure, and payment history
Cleaning Practice	The dataset may include missing values and one duplicate row, so students should practice data preparation before clustering

Target variable

Renewed_Membership is not used as a clustering input. After clusters are found, students should use it to check whether different clusters show different renewal rates, which validates that the segments are meaningful for the business.

Input features

Students should use the numerical customer and behavioral variables as input features for clustering. Categorical features should be excluded from K-Means input.

- Subscription behavior: Months_As_Customer, Monthly_Orders, Avg_Order_Value
- App engagement: App_Sessions_Per_Month, Last_Login_Days
- Service quality: Late_Deliveries_Last_3M, Delivery_Rating, Support_Tickets_Last_3M
- Customer profile: Discount_Usage_Rate, Household_Size, Payment_Issues_Last_3M

5. Why This Analysis Is Useful

Discovering customer segments turns behavioral data into actionable business strategy. A marketing manager can use cluster profiles to decide which customers need service recovery, which customers deserve loyalty rewards, and which customers need re-engagement campaigns.

- Better targeting: focus retention spend on the customer groups most at risk of churning.
- Budget control: avoid wasting discount offers on customers who are already likely to renew.
- Service improvement: identify which customer groups experience the worst delivery quality.
- Customer experience: deliver the right message to the right customer at the right time.
- Business communication: explain cluster profiles to non-technical managers using interpretable segment labels.

6. Evaluation Approach for the Business

For unsupervised clustering, the Elbow Method is the primary model selection tool. It measures Within-Cluster Sum of Squares (inertia) for different values of K and helps select the point where adding more clusters provides diminishing returns. After clustering, the business should evaluate each segment using cluster means and renewal rate analysis.

Method	What It Measures	Business Interpretation
Elbow Method	Inertia (within-cluster sum of squares) for K=1 to K=10	Helps select the optimal number of clusters by finding the bend point where inertia reduction slows significantly.
Cluster Means	Average value of each feature within each cluster	Reveals the behavioral profile of each segment, allowing managers to assign meaningful labels.
Cluster Size	Number of customers assigned to each cluster	Ensures segments are large enough to act on. Very small clusters may indicate outliers.
Renewal Rate by Cluster	Percentage of Renewed_Membership = Yes within each cluster	Validates that clusters correspond to meaningful differences in real business retention outcomes.

7. Cluster Profiles Found

K = 4 was selected using the Elbow Method. The analysis revealed four distinct customer segments with meaningfully different behavioral profiles and renewal rates.

Cluster	Segment Label	Key Characteristics
Cluster 0	Passive Low-Spenders (134 customers)	Low orders (2.7/mo), lowest order value ($46.73), fewest app sessions (11). Very few delivery issues. Lowest renewal rate (~10%).
Cluster 1	Active At-Risk (81 customers)	High app sessions (22/mo), moderate orders (5.3/mo), but highest late deliveries (1.56) and most support tickets (1.11). Lowest delivery rating (3.55). ~15% renewal rate.
Cluster 2	High-Value Loyalists (121 customers)	Highest orders (6.6/mo), highest order value ($61.70), most app sessions (26/mo). Fewest late deliveries, best ratings (4.25). Highest renewal rate (~47%).

Cluster 3	Payment-Friction Customers (24 customers)	Moderate engagement (4.1 orders, 17 sessions/mo). Only cluster with payment issues (avg 1.0). ~17% renewal rate.

8. Expected Important Features

After creating the clusters, students should examine which features drive the separation between groups. The most important distinguishing features may vary depending on preprocessing choices and the number of clusters selected. However, the following variables are expected to be meaningful from a business perspective:

- App_Sessions_Per_Month: customers who engage more with the app are more likely to see value in their subscription and renew.
- Monthly_Orders and Avg_Order_Value: order frequency and spending level reflect how much value a customer extracts from the membership.
- Late_Deliveries_Last_3M: poor delivery performance is a key driver of dissatisfaction for engaged customers.
- Delivery_Rating: directly measures customer satisfaction with the core service.
- Support_Tickets_Last_3M: high complaint volume indicates service problems that may push customers toward churning.
- Payment_Issues_Last_3M: payment friction is a distinct barrier to renewal that separates one cluster entirely.
- Last_Login_Days: recency of engagement indicates whether a customer is still actively using the platform.

9. How the Business Can Use the Results

The clustering results should be translated into practical retention strategies. Students should connect cluster profiles and renewal rates to specific marketing actions rather than only reporting technical outputs.

Cluster Insight	Possible Business Action
Cluster 2 has the highest renewal rate and highest spending	Protect with exclusive loyalty rewards and early access to new products. Consider upselling to premium membership tiers.
Cluster 1 has high engagement but worst delivery experience	Launch service recovery outreach before renewal. Acknowledge delivery issues proactively and offer a targeted follow-up offer after fixing the problem.
Cluster 0 has very low engagement and lowest renewal rate	Send re-engagement campaigns featuring underused app features. Offer a reduced-cost membership tier or a trial extension.
Cluster 3 is the only group with payment issues	Proactively contact these customers about payment barriers. Offer flexible billing options, alternative payment methods, or a short grace period.

10. Limitations and Possible Bias

This dataset is simplified for learning purposes. It may not include every factor that influences real customer churn behavior. The model may also learn patterns from incomplete, imbalanced, or biased data.

- The dataset does not include external context such as household income, local competition, or personal life events that may cause churn unrelated to service quality.

- K-Means assumes spherical, equally-sized clusters and uses Euclidean distance, which may not perfectly reflect how customers naturally group in real data.
- Categorical features (Age_Group, City_Type, Membership_Type) were excluded from clustering, which may limit the richness of the segments found.
- Cluster 3 is very small (24 customers), which may make its boundaries less stable across different random initializations.
- Demographic variables may introduce fairness concerns if used to make automated decisions without human review.

11. Why Human Judgment Is Still Needed

Human judgment is necessary because retention decisions affect real customers, marketing budgets, and brand reputation. A model can support decisions, but managers must interpret results carefully, check for bias, consider customer experience, and make decisions that align with business strategy and ethical standards.

- Managers should review whether the cluster profiles and recommended actions make business sense in the current operational context.
- Humans should check whether the clustering model might inadvertently disadvantage certain customer groups based on geography or demographics.
- Marketing teams should respect customer privacy and avoid excessive or intrusive targeting based on model outputs.
- Service improvements should be made for at-risk clusters rather than simply predicting churn and accepting it as inevitable.
- The final decision should combine model results, business goals, frontline knowledge, and responsible human oversight.

12. Suggested Student Interpretation Focus

In the assignment notebook, students should not only show code outputs. A strong submission should explain results in business language and connect cluster profiles, renewal rates, and feature means to practical business decisions.

- Explain the business problem and why customer segmentation is useful before building the model.
- Clearly identify which numerical features were selected for clustering and why categorical features were excluded.
- Use clear preprocessing steps: median imputation for numerical features, mode imputation for categorical features, and StandardScaler before K-Means.
- Use the Elbow Method to justify the number of clusters selected and explain the reasoning in plain language.
- Interpret cluster mean tables as business profiles, not only as numerical outputs.
- Connect renewal rates by cluster to the business problem of subscription churn.
- End with one limitation of the dataset or model and one responsible AI reflection.